Negative values and signed operations in photonic neural networks

Why it is difficult in Photonic Neural networks to represent negative values. And how singed matrix vector multiplication is implemented in optical domain?

Marshaled technical expertise on photonic neural networks

Negative Values in Photonic Neural Networks

Why It's Fundamentally Difficult

The core problem is **physical**: optical intensity (power) is strictly non-negative. Light cannot carry a negative intensity — a photodetector measures photon flux, which is always ⩾ 0. This means the natural "currency" of photonic computing is inherently unsigned.

Specifically:

- **Photodetectors** output a current proportional to optical power $P \propto |E|^2 \geq 0$
- **Modulators and weight banks** (MZIs, ring resonators, VOAs) control *attenuation*, i.e., they can only scale a signal between 0 and 1 — they cannot natively produce a sign flip
- **Matrix-vector multiplication** in optics is naturally implemented as a sum of products of intensities, all positive

This is a fundamental mismatch with neural network weights, which are real-valued and typically distributed around zero (roughly half negative, half positive).

Strategies for Signed Matrix-Vector Multiplication

Several techniques have been developed to encode and compute with signed weights:

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1. 🔵 Differential (Balanced) Detection

The most common and hardware-efficient approach.

Idea: Represent a signed weight w as the *difference* of two non-negative values:

$$w = w^+ - w^-, \quad w^+, w^- \geq 0$$

Implementation:

- Split each weight into a positive part w^+ and a negative part w^-, implemented on **two separate photonic paths**
- Each path performs a non-negative analog multiply-accumulate
- A **balanced photodetector pair** subtracts the two photocurrents:

$$i_{out} = R(P^+ - P^-) \propto w^+x - w^-x = wx$$

Cost: Doubles the hardware (2× MZIs, 2× detectors per weight). Used in systems like the MIT/Lightmatter approach.

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2. 🟠 Differential MZI Encoding

In **Mach-Zehnder Interferometer (MZI)** mesh architectures (e.g., Clements or Reck decomposition):

- An MZI naturally splits power between two output ports
- The **two output ports are complementary**: if one carries transmission T, the other carries $1 - T$
- By taking the *difference* of the two output port signals, you recover a signed weight centered at zero:

$$w \propto T - (1 - T) = 2T - 1 \in [-1, +1]$$

This is elegant because the sign comes for free from the MZI's inherent 2-port nature.

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3. 🟡 Optical Field (Amplitude) Encoding — Coherent Systems

Instead of encoding information in **intensity**, use the **electric field amplitude**, which *can* be negative (or complex):

$$E(t) = A \cdot e^{i\phi}$$

- Use **coherent light** with a phase reference (local oscillator)
- Weights are encoded in the **field amplitude** A (which is bipolar in the complex sense)
- Interference between signal and reference allows subtraction
- **Coherent detection** (homodyne/heterodyne) recovers the signed value

Downside: Requires phase stability, coherent sources, and is sensitive to thermal/mechanical noise. Much harder to integrate at scale.

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4. 🟢 Time-Division / WDM Differential Scheme

Used in **broadcast-and-weight** architectures:

- Positive and negative weights are assigned to **different time slots** or

wavelength channels

- Partial sums are accumulated separately, then electronically subtracted at the end
- Trades optical complexity for electronic post-processing

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5. 🔴 Bias Shifting (Offset Method)

A simpler but less efficient trick:

- Shift all weights by a constant offset b so they become non-negative: $w' = w + b \geq 0$
- After optical computation, subtract the bias contribution electronically:

$$y = W'x - bx = Wx$$

Downside: Reduces the dynamic range available for actual weight variation; the bias consumes headroom.

Summary Table

Method	Signed Range	Hardware Cost	Noise Sensitivity	Used In
Differential detection	Full $[-1, +1]$	2× components	Moderate	Lightmatter, Luminous
Differential MZI ports	Full $[-1, +1]$	None extra	Low	Shen et al. (MIT)
Coherent field encoding	Full complex	High	Very High	Research systems
WDM/Time differential	Full $[-1, +1]$	Moderate	Low	Broadcast architectures
Bias shifting	Reduced	Minimal	Low	Simple prototypes

Bottom Line

The **differential MZI** and **balanced detection** approaches are the dominant practical solutions. The inability to natively represent negative values is one of the key reasons photonic neural networks have higher hardware overhead than their electronic counterparts for equivalent model capacity — you're essentially paying a ~2× cost in devices to recover the full signed number representation that electronics gets for free.

I want to give me a well mathematical grounded reasone why representing negative values are difficult in optical domain?

Thinking about mathematical foundations of negative value representation in optical systems >

Here is the mathematical foundation for why negative values are fundamentally problematic in photonics.

The issue traces back to the physics of how we *detect* and *measure* light, which is governed by laws that enforce non-negativity at a deep level.

The Physics: Why Optical Intensity is Always ⩾ 0

An electromagnetic wave propagating in a medium has an electric field:

$$E(r,t) = A \cdot e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$$

The physically measurable quantity — what a photodetector sees — is the **optical intensity**, defined as the time-averaged Poynting vector magnitude:

$$I = \langle S \rangle = \frac{1}{2}\varepsilon_0 c|E|^2 = \frac{1}{2}\varepsilon_0 c \cdot A^2 \geq 0$$

This is a squared modulus. **Squaring is a non-negative operation by definition**, so $I \geq 0$ for all possible field amplitudes $A \in \mathbb{C}$. There is no physical configuration of light that yields a negative intensity.

The Measurement Problem: Photodetector Response

A photodetector converts photons to electrons via the photoelectric effect. The generated photocurrent is:

$$i_{ph} = \mathcal{R} \cdot P_{opt} = \mathcal{R} \cdot \int_{\mathcal{A}} I \, dA \geq 0$$

where $\mathcal{R}$ [A/W] is the responsivity and P_{opt} is the optical power. Since $I \geq 0$ everywhere on the detector area $\mathcal{A}$, the integral is non-negative, and therefore $i_{ph} \geq 0$ **always**. The output of any photodetector is a non-negative current — it cannot produce a negative photocurrent.

The Linear Algebra Consequence

In a Photonic Neural Network, a Matrix-Vector Multiplication (MVM) is performed optically. An input vector $\mathbf{x} \in \mathbb{R}^n$ is encoded as a vector of optical intensities:

$$\mathbf{x} \mapsto \mathbf{P} = [P_1, P_2, \ldots, P_n], \quad P_i \geq 0$$

A weight matrix W acts on this via the optical system (e.g. an MZI mesh). The output at each detector is:

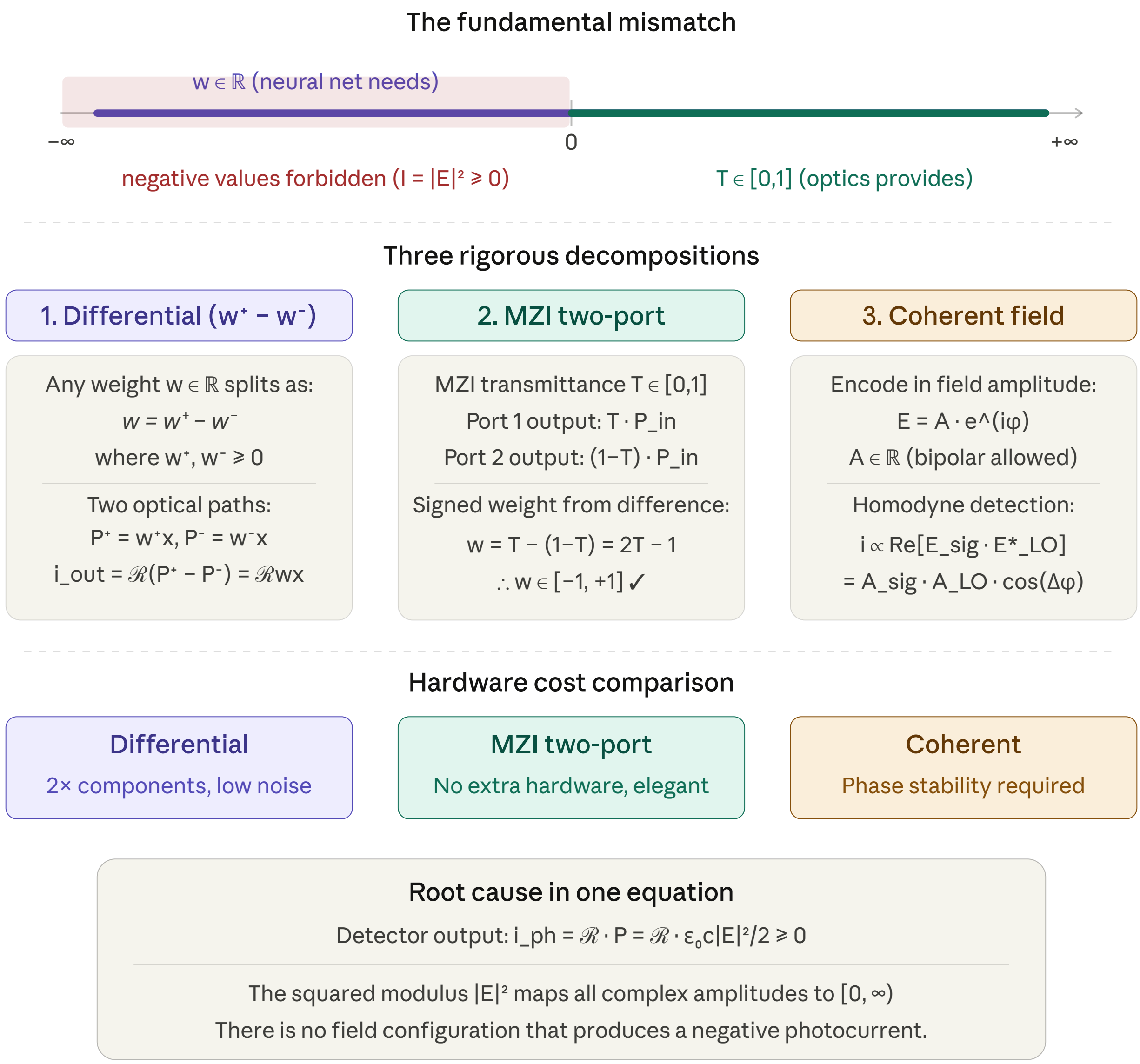
$$y_j = \sum_{i=1}^{n} T_{ji} \cdot P_i$$

where $T_{ji} \geq 0$ is the **power transmittance** from input port i to output port j. Transmittance is a ratio of powers, so:

$$0 \leq T_{ji} \leq 1 \quad \forall i, j$$

This means the optical system naturally computes only **non-negative matrix-vector products**. The weight matrix W is constrained to $W \in [0, 1]^{m \times n}$. Neural network weight matrices, however, have entries $w_{ji} \in \mathbb{R}$ — approximately half are negative — so the optical domain can only represent *half* the required weight space.

Now let's look at what the diagram below illustrates — the mismatch between the signed weight space neural nets need versus the unsigned space optics naturally provides, and then the three rigorous decompositions that solve it.



Why the Decompositions Work Mathematically

Decomposition 1 — Jordan Decomposition (Differential)

Any real number w can be uniquely decomposed into its **positive and negative parts**:

$$w = \underset{w^+ \geq 0}{\max(w, 0)} - \underset{w^- \geq 0}{\max(-w, 0)}$$

This is the **Jordan decomposition** of a signed measure. Optically, w^+ and w^- are encoded as two separate transmittances, and balanced photodetection subtracts the resulting photocurrents. The cost is exactly 2× hardware.

Decomposition 2 — Affine Mapping via MZI Port Identity

An MZI has transmittance $T \in [0, 1]$ to port 1 and $(1 - T)$ to port 2 by energy conservation (lossless beamsplitter). Their difference gives an **affine map** from $[0, 1]$ onto $[-1, +1]$:

$$w = T - (1 - T) = 2T - 1, \quad T = \frac{w + 1}{2}$$

This is essentially shifting the origin of the unsigned domain. No extra hardware — the second port already exists physically.

Decomposition 3 — Field Encoding (Coherent Detection)

Homodyne detection measures the **real part of the cross-correlation** between signal and local oscillator fields:

$$i \propto \mathrm{Re}[E_{sig} \cdot E_{LO}^*] = A_{sig} A_{LO} \cos(\Delta\phi)$$

Since $\cos(\Delta\phi) \in [-1, +1]$, the photocurrent is now genuinely bipolar. The "squaring" problem is bypassed because detection happens *before* squaring — but this requires a phase-stable reference, which is the key engineering challenge.

Summary in One Line

The obstacle is purely mathematical: **photodetection is a map $\mathbb{C} \to \mathbb{R}_{\geq 0}$ via the squared modulus $|E|^2$**, and no invertible extension of this map can produce negative outputs. All strategies for signed computation must work *around* this squaring operation — either by decomposing signed weights into two unsigned quantities (differential), remapping the domain (MZI port), or circumventing the squaring entirely via coherent measurement.

What's the difference between Coherent and Incoherent photonics?

Orchestrated visual aid and prepared comprehensive mathematical analysis >

The difference runs deeper than just a design choice — it's about which physical quantity carries information. Let me build this up from the wave equation level.

Incoherent

Information carrier
I = |E|² (intensity, phase lost)

Random phases — intensities add
I_total = Σ|E_i|² (no cross-terms)

Detector output

i_ph = ℛ · |E|²
Phase destroyed at detection
output ∈ [0, +∞)

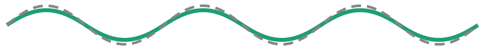
0
Weights ∈ [0, 1] — signed needs 2× hw

System properties

Phase stability	Not required
Native weight range	[0, 1] unsigned
Signed MVM	Differential detection needed
Fabrication	Simple, robust, scalable
Dominant noise	Shot + thermal noise
Current PNN maturity	Most systems today

Coherent

Information carrier
E = A · e^(iφ) (amplitude + phase)



Signal Local oscillator (LO)
Fixed phase — fields add: E_out = E_s + E_LO

Detector output

i = ℛ · Re[E_s · E*_LO]
= ℛ · A_s · A_LO · cos(Δφ) signed
i < 0 i > 0

0
Weights ∈ [−1, +1] — signed is native

System properties

Phase stability	Critical — must lock phase
Native weight range	[−1, +1] signed
Signed MVM	Native — homodyne detection
Fabrication	Complex, temp-sensitive
Dominant noise	Shot-noise limited
Current PNN maturity	Research / emerging

The Core Mathematical Distinction

The entire difference reduces to *which term of the electromagnetic field equation carries information*.

Incoherent: Information in Intensity

In an incoherent system, light from multiple independent sources (or multiple modes) is used. Each source k has a random, time-varying phase $\phi_k(t)$. The total field is:

$$E_{total}(t) = \sum_k A_k \, e^{i(\omega t + \phi_k(t))}$$

When the detector integrates over its response time $\tau_{det} \gg \tau_{coherence}$, the cross terms average to zero:

$$I_{total} = \langle |E_{total}|^2 \rangle = \sum_k |A_k|^2 + \underbrace{\sum_{j \neq k} \left\langle A_j A_k \, e^{i(\phi_j - \phi_k)} \right\rangle}_{\to 0 \text{ (random phases)}}$$

The cross-term vanishes because the phases are uncorrelated: $\langle e^{i(\phi_j - \phi_k)} \rangle = 0$. **Intensities simply add.** The only observable is $I \geq 0$.

Coherent: Information in the Full Field

In a coherent system, a single-frequency laser with a stable phase is split into a signal E_s and a local oscillator E_{LO}. They share the same $\phi(t)$, so the cross-term survives:

$$I = |E_s + E_{LO}|^2 = \underbrace{|E_s|^2}_{DC} + \underbrace{|E_{LO}|^2}_{DC} + \underbrace{2\,\mathrm{Re}[E_s \cdot E_{LO}^*]}_{\text{interference term — can be negative!}}$$

Homodyne detection subtracts the two output ports of a beam splitter, canceling the DC terms and leaving only:

$$i_{hom} \propto 2\,\mathrm{Re}[E_s \cdot E_{LO}^*] = 2A_s A_{LO} \cos(\Delta\phi)$$

Since $\cos(\Delta\phi) \in [-1, +1]$, the photocurrent is **genuinely signed**. The sign is controlled by the relative phase $\Delta\phi$, and flipping $\Delta\phi \to \Delta\phi + \pi$ negates the output.

The Superposition Principle Formulated

This is really the difference between **incoherent superposition** and **coherent superposition**:

	Rule	Result
Incoherent	$I_{total} = \sum_k I_k$	Powers add, no interference
Coherent	$E_{total} = \sum_k E_k$, then $I =	E_{total}

The cross-term $2\mathrm{Re}[E_1 E_2^*]$ is the mathematical engine behind all signed computation in photonics. It exists only when the two fields have a well-defined, stable phase relationship — i.e., when they are mutually coherent.

Why Coherence Is Hard to Maintain in Practice

The phase $\phi(t)$ of a laser drifts due to quantum phase noise (linewidth), thermal expansion of waveguides (~1 pm/°C for silicon), mechanical vibration, and carrier-induced index changes. The coherence time is:

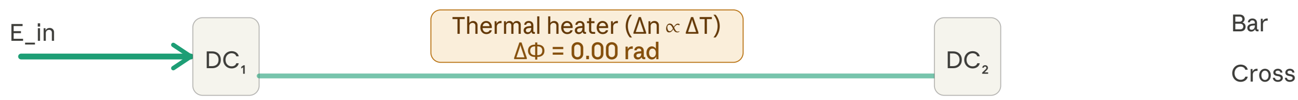
$$\tau_c = \frac{1}{\pi \Delta \nu}$$

For a typical DFB laser with linewidth $\Delta\nu \approx 1$ MHz, $\tau_c \approx 300$ ns. This means the phase relationship between signal and LO is only stable over a few hundred nanoseconds — the entire computation must complete within τ_c, or the cross-term washes out and you get incoherent behavior by accident. This is why incoherent systems, while restricted to non-negative weights, are far more practically robust in today's PNN hardware.

Is thermal shifted MZIs are incoherent or coherent? Especially under the framework of photonic neural networks

Architected interactive MZI visualization and coherence classification framework >

Great question — this is one of the most commonly misunderstood points in PNN literature. The short answer is: thermal MZIs are **coherent in mechanism** but occupy a precise middle ground called *self-coherent*. Let me explain why.



DC_2 coherently combines the fields: I_bar = |E_upper + E_lower|² = $\cos^2(\Delta\Phi/2) \cdot P_{in}$

W = T_bar − T_cross = $\cos(\Delta\Phi) \in [-1, +1]$ signed weight — self-coherent, no external LO needed

$\Delta\Phi = 0$ ◯ 2π

1.000	0.000	+1.000	Constructive
T_bar = $\cos^2(\Delta\Phi/2)$	T_cross = $\sin^2(\Delta\Phi/2)$	W = T_bar − T_cross = $\cos(\Delta\Phi)$	Interference mode

Classification — self-coherent: The MZI requires field interference at DC_2, so it demands a coherent (single-mode) source. But unlike homodyne detection there is no external LO — the two arms originate from the same photon and serve as each other's phase reference. The signed weight W = $\cos(\Delta\Phi) \in [-1,+1]$ emerges from differential port readout within this self-referencing structure.

The Precise Answer: Self-Coherent

A thermal MZI is **coherent in its physical mechanism** for a fundamental reason: DC_2 (the output directional coupler) computes a *field sum*, not an *intensity sum*. This means the phase of the field in both arms is what determines where the power goes:

$$I_{bar} = \left. \frac{E_{in}}{2} + \frac{E_{in} \cdot e^{i\Delta\Phi}}{2} \right.^{2} = P_{in} \cos^2\left(\frac{\Delta\Phi}{2}\right)$$

If you set $\Delta\Phi = \pi$, the two fields *cancel* at the bar port — that is **destructive interference**. Destructive interference is a coherent phenomenon. It simply cannot happen in an incoherent system, where $I_{total} = I_1 + I_2 > 0$ always.

Why It's Not Fully Coherent Either

But it also falls short of the "full homodyne coherent" definition, for a precise structural reason:

Property	Incoherent	Thermal MZI	Full homodyne
Source requirement	Any (broadband OK)	Single-mode laser	Narrow-linewidth, phase-locked laser
Reference signal	None	Same photon (self-referenced)	Separate LO laser
Phase info at output	Destroyed	Preserved *within* the device	Fully recovered at detector
Signed weights native	No	Yes, via $W = \cos(\Delta\Phi)$	Yes, via $i \propto \cos(\Delta\phi_{LO})$
Sensitivity to vibration/temp	Low	Moderate	Very high

The critical distinction is **where the phase reference comes from**. In an MZI, the two interfering arms are both derived from the *same input field* — DC_1 splits E_{in} into two copies. The heater shifts the phase of one copy relative to the other. DC_2 then recombines them. No external laser reference is needed; the MZI is *self-referencing*.

In contrast, a homodyne system requires a *separate* LO laser whose phase is actively locked to the signal laser. That is far more demanding.

The MZI Transfer Matrix (Why Phase = Weight)

The full MZI is a cascade of three unitary operators:

$$U_{MZI} = U_{DC} \cdot \begin{pmatrix} e^{i\Delta\Phi} & 0 \\ 0 & 1 \end{pmatrix} \cdot U_{DC}$$

where $U_{DC} = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ is the 3dB directional coupler. Working this out:

$$U_{MZI} = \begin{pmatrix} i\sin(\Delta\Phi/2) & \cos(\Delta\Phi/2) \\ \cos(\Delta\Phi/2) & -i\sin(\Delta\Phi/2) \end{pmatrix} e^{i\Delta\Phi/2}$$

So the field amplitudes at the two ports are:

$$E_{bar} = \cos\left(\tfrac{\Delta\Phi}{2}\right) E_{in}, \qquad E_{cross} = i\sin\left(\tfrac{\Delta\Phi}{2}\right) E_{in}$$

Detecting both ports and taking their difference gives a **signed weight with no extra hardware**:

$$W = |E_{bar}|^2 - |E_{cross}|^2 = \cos^2\left(\tfrac{\Delta\Phi}{2}\right) - \sin^2\left(\tfrac{\Delta\Phi}{2}\right) = \cos(\Delta\Phi) \in [-1, +1]$$

This is the reason thermal MZI PNNs (Shen et al. 2017, Lightmatter, etc.) can implement fully signed matrix-vector multiplication. The thermal heater sets $\Delta\Phi$, the differential photodetector computes $W = \cos(\Delta\Phi)$, and the weight naturally spans the full signed range as $\Delta\Phi$ sweeps from 0 to π.

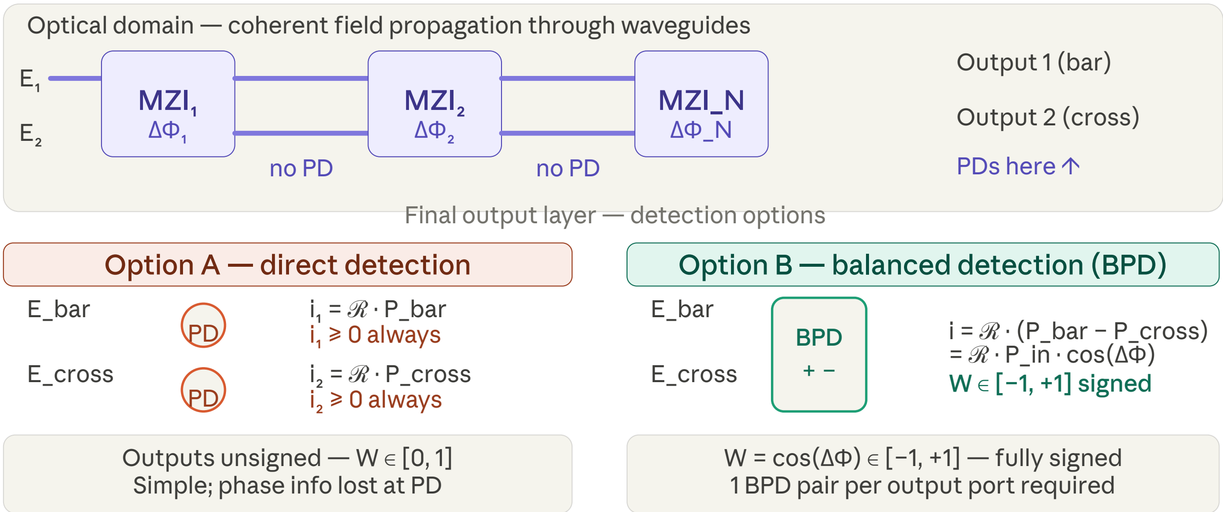
Practical Implication for PNN Design

Because the MZI is self-coherent (not full homodyne), the coherence length requirement is only that the laser's coherence time exceeds the path length difference *between the two arms* — typically a few hundred microns on-chip, which is easily satisfied by any DFB or VCSEL laser. You don't need to lock a separate LO. This is what makes thermal MZI meshes the dominant PNN architecture today: they get the signed weight benefit of coherent computation without the stabilisation complexity of full homodyne systems.

However, how the values are read after in both cross and bar? There no photodetector?

Architected comprehensive explanation of photodetector placement in MZI meshes >

Excellent practical question — and it reveals a key architectural detail that most PNN papers gloss over. The answer is: **there are no photodetectors at intermediate MZI ports**. The bar and cross ports within the mesh are just waveguide connections. PDs only appear at the final output layer.



Why There Are No PDs Between MZIs

Within the mesh, the bar and cross ports of one MZI are simply **waveguides** that feed directly into the input ports of the next MZI. Putting a photodetector there would terminate the optical signal — you'd destroy the field and replace it with a photocurrent, ending the computation. The whole point of the optical mesh is to let the field propagate and accumulate transformations *without detecting it*.

The MZI mesh implements a **unitary matrix** U on the input field vector $\mathbf{E}_{in}$. This happens entirely in the optical domain:

$$\mathbf{E}_{out} = U\,\mathbf{E}_{in}$$

where U is decomposed into a product of 2×2 MZI rotation matrices. The field just propagates through — no readout happens until the very end.

The Two Detection Choices at the Output

Option A — direct detection: Each output port gets a single photodetector. This measures intensity $i_k = \mathcal{R} \cdot |E_k|^2 \geq 0$. Simple and robust, but you lose all phase information and weights are effectively unsigned. Most practical deployed systems use this.

Option B — balanced detection (BPD): A pair of output ports feeds into a *balanced photodetector*, which subtracts the two photocurrents electronically:

$$i_{diff} = \mathcal{R}(P_{bar} - P_{cross}) = \mathcal{R} \cdot P_{in} \cdot \cos(\Delta\Phi)$$

This recovers the signed weight $W = \cos(\Delta\Phi) \in [-1, +1]$ discussed earlier. The subtraction happens *electronically inside the BPD chip* — no optical phase reference needed.

The Multilayer Problem

This raises a deeper issue for deep PNNs. A single MZI mesh does one linear transformation. A neural network needs alternating **linear layers and nonlinear activations** (ReLU, sigmoid, etc.). After the first optical layer, you must:

1. **Detect** the output (PDs) → converts field to current
2. **Apply nonlinearity** electronically (ADC → ReLU → DAC)
3. **Re-encode** back into optical domain (modulators)
4. **Pass through the next MZI mesh**

This is called the **optoelectronic (OE) approach** and is used in systems like Lightmatter's Envise. Each boundary between layers incurs an optical→electronic→optical conversion penalty, which is one of the central engineering challenges in scaling PNNs to many layers.

How to export this conversation and keep ti locally?

Identified user's intent to locally archive conversation >

Here are the main ways to save this conversation locally:

Built-in browser options (easiest)

- **Print to PDF:** Press `Ctrl+P` (Windows/Linux) or `Cmd+P` (Mac) in your browser, then choose "Save as PDF" as the destination. This preserves the diagrams and formatting exactly as you see them.
- **Save as HTML:** `Ctrl+S` / `Cmd+S` saves the full page as an HTML file (with a folder of assets). Works but can be bulky.

Claude.ai built-in (if available)

- Look for a **"Share" or export button** in the top-right of the conversation — depending on your plan, there may be a direct export option. I'd recommend checking, though I'm not certain of the exact current UI.

Copy-paste to a document

- Select all (`Ctrl+A` / `Cmd+A`) on the page, copy, and paste into a Word doc or Notion page. The SVG diagrams won't transfer well this way, but the text will be clean.

Browser extensions

- Extensions like **MarkDownload** (Chrome/Firefox) can save the page as a clean Markdown `.md` file, which is good for the text content.

For a conversation this technical with the diagrams, **Print to PDF** is almost certainly your best option — it captures everything in one clean file. The interactive slider in the MZI widget won't work in PDF, but everything else renders well.

Give me a beamer frame in latex with two columns. One Has the text "
Incoherent Photonics face challenges in supporting negative quantities, since optical
 signals are naturally converted into power signals during the non-linear
 process that must occur at the activation stage, implying that the sign
 information of the weighted sum is ignored.
" and the other the nice graph on incoherent with the wave and the positive axis as well as the explanation why the Intesity is always > 0

Architected two-column Beamer frame with text and visualization >

The file is ready. A few notes on dependencies in your preamble:

latex

```latex
\usepackage{tikz}
\usetikzlibrary{arrows.meta}
```

That's all it needs — the frame uses only `tikzpicture` for the graph and standard Beamer `columns` / `block` environments. The TikZ plot draws $E(t) = \sin(\omega t)$ in blue and $I(t) = |E(t)|^2 = \sin^2(\omega t)$ in red, with a shaded forbidden region below zero and a small summary table at the bottom right.

Incoherent frame
TEX

Download